

Anora: SE Experiment Documentation

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1. Requirements

a) Functional Requirements

These requirements describe the features and functionalities that the Anora platform must offer to ensure privacy-first mental wellness tracking.

1.1 Secure Journaling System

- **FR1.1.1:** The system must allow users to create text-based journal entries.
- **FR1.1.2:** All journal entries must be encrypted using AES-256 before being stored in the local database (SQLite/Hive).
- **FR1.1.3:** The system must ensure no raw text is ever transmitted to a cloud server.

1.2 On-Device AI Analysis

- **FR1.2.1:** The system must utilize a quantized MentalBERT model to perform real-time emotion analysis on journal entries.
- **FR1.2.2:** The platform must detect risk flags (e.g., Self-harm, Anxiety, Depression) locally without internet access.

1.3 Clinician "Locked Box" Reporting

- **FR1.3.1:** Users must be able to generate a "Locked Box" export containing a summary of insights.
- **FR1.3.2:** The system must encrypt the export using a one-time AES key, which is then encrypted with the clinician's public RSA key.

1.4 Federated Learning Integration

- **FR1.4.1:** The system must calculate model gradients locally during device idle time.
- **FR1.4.2:** The platform must mask these gradients using Secure Aggregation (SecAgg) before sending them to the backend for global model updates.

1.5 Privacy Guard & Analytics

- **FR1.5.1:** The system must include a "Privacy Guard" layer that blocks any HTTP request containing raw text strings.
- **FR1.5.2:** The dashboard must display derived analytics (mood trends, sleep alignment) without exposing underlying raw data.

2. Sequence of Phases

- **Communication:** In the Communication phase, deliverables include the **Privacy & Requirements Specification**, which defines the "Zero-Knowledge" architecture and stakeholder expectations (Patient vs. Clinician).
- **Planning:** During the Planning phase, the deliverables consist of a **Project Timeline** and a **Threat Model Risk Assessment**. This includes analyzing vectors for data leakage and defining the "Blind Mailman" backend scope.
- **Modeling:** In the Modeling phase, the main deliverables are the **System Architecture Diagram** (demonstrating the Local-First approach) and **Data Flow Diagrams** (visualizing the Red/Green/Blue data classification).
- **Construction:** The Construction phase produces the **Source Code** (Flutter for mobile, Python/FastAPI for backend). Unit Tests are created for the encryption modules, and Integration Tests ensure the TFLite model runs correctly on-device.
- **Deployment:** In the Deployment phase, the plan includes distributing the Flutter app to app stores and deploying the Dockerized backend. User manuals emphasize key management (e.g., "If you lose your device, data is lost" due to zero-knowledge).

3. Incremental Approach

- **Increment 1:** Develop the core Flutter UI and local SQLite database with basic AES-256 encryption implementation.
- **Increment 2:** Integrate the TFLite inference engine and the quantized MentalBERT model for local sentiment analysis.
- **Increment 3:** Implement the "Privacy Guard" network interceptor and the local analytics dashboard.
- **Increment 4:** Develop the "Locked Box" cryptography module (RSA key exchange) for clinician reporting.
- **Increment 5:** Build the "Blind Mailman" backend using Python/FastAPI to handle encrypted blobs.
- **Increment 6:** Implement Federated Learning clients to compute and mask gradients locally.
- **Increment 7:** Finalize the Clinician Dashboard for decrypting and viewing reports.

4. Iterative Model

Phase 1: Planning Define the iteration's goal (e.g., "Enable secure reporting"). Prioritize features based on the "Local-First" principle.

Phase 2: Analysis and Design Refine the cryptographic protocols for the specific iteration. Update the Data Journey Diagram to reflect new data paths (e.g., adding the Federated Learning loop).

Phase 3: Implementation Code the features in Dart (Flutter) and Python. Optimize the TFLite model for latency (<200ms).

Phase 4: Testing and Evaluation Validate that no raw text leaks during the new features' operation. Conduct performance testing on low-end devices.

Specific Iterations

Iteration 1: Core Journaling & Encryption (3 Weeks)

- **Goals:** Implement secure local storage, basic journaling UI, and AES encryption.
- **Outcome:** A functioning diary app where data is unreadable outside the app context.

Iteration 2: On-Device AI Engine (3 Weeks)

- **Goals:** Integrate MentalBERT (TFLite), implement tokenization, and generate mood scores.
- **Outcome:** Real-time emotional analysis running entirely offline.

Iteration 3: The Care Loop (Reporting) (3 Weeks)

- **Goals:** Implement RSA/AES hybrid encryption, the "Locked Box" packaging, and the backend "Blind Mailman" API.
- **Outcome:** Users can safely share summaries with doctors without the server seeing the data.

Iteration 4: Federated Learning (3 Weeks)

- **Goals:** Implement local gradient calculation and the Secure Aggregation protocol.
- **Outcome:** The model improves over time without collecting user data.

Iteration 5: Analytics & Polish (3 Weeks)

- **Goals:** specific dashboards for risk trends, UI polish, and optimization of app size (<50MB).
- **Outcome:** A production-ready, clinical-grade application.

5. Spiral Model Plan for Anora

Spiral 1: Basic Privacy & Journaling (7 Weeks)

- **Goal:** Establish the "Zero-Knowledge" storage foundation and basic UI.
- **Planning:** Define the encryption standards (AES-256) and database schema (Hive/SQLite).
- **Risk Analysis:**
 - *Risk:* Encryption causing UI lag.

- *Mitigation:* Test flutter_secure_storage performance early.
- **Engineering:** Build the Flutter app, local storage, and key management via Secure Enclave.
- **Evaluation:** Verify that database files are unreadable when extracted from the device.

Spiral 2: Intelligent Insights (6 Weeks)

- **Goal:** Enable on-device AI inference without internet.
- **Planning:** Select the MentalBERT model and define quantization strategy (INT8).
- **Risk Analysis:**
 - *Risk:* Model size exceeding mobile limits (>100MB).
 - *Mitigation:* Aggressive quantization and pruning during conversion to TFLite.
- **Engineering:** Integrate TensorFlow Lite, build the inference pipeline, and design the Mood/Risk dashboard.
- **Evaluation:** Test inference accuracy against test datasets; ensure latency is under 200ms.

Spiral 3: Connected Ecosystem (8 Weeks)

- **Goal:** Implement the "Locked Box" sharing and Federated Learning.
- **Planning:** Design the "Blind Mailman" API and SecAgg protocol.
- **Risk Analysis:**
 - *Risk:* Man-in-the-Middle attacks during key exchange.
 - *Mitigation:* Strict certificate pinning and RSA-2048 implementation.
- **Engineering:** Develop the Python backend, implement the masking protocol for gradients, and build the clinician decryption tool.
- **Evaluation:** Conduct a security audit to ensure no raw data leaks during transmission.

6. Waterfall Model

- 1. Requirements Analysis (1 Month)** Gather clinical requirements for DSM-5 alignment and privacy constraints (GDPR/HIPAA compliance) for the Zero-Knowledge architecture.
- 2. System Design (2 Months)** Create detailed architectural diagrams for the "Local-First" logic, including the "Privacy Guard" firewall and "Locked Box" cryptography flows.
- 3. Implementation (3 Months)** Develop the Flutter mobile app, train/quantize the MentalBERT model, and build the FastAPI backend.
- 4. Integration and Testing (2 Months)** Perform rigorous security testing (penetration testing) on the "Blind Mailman" and functional testing on the AI inference accuracy.
- 5. Deployment (1 Month)** Release to App Store/Play Store, deploy backend containers, and publish clinician onboarding guides.

6. Maintenance (Ongoing) Monitor model drift (via Federated Learning metrics) and update security libraries.

Key Documentation

- **Requirements:** Privacy Spec, DSM-5 Mapping.
- **Design:** Cryptographic Protocol Spec, TFLite Model Card.
- **Implementation:** API Documentation (Blind Mailman), DartDocs.
- **Testing:** Penetration Test Report, AI Accuracy Report.
- **Deployment:** Clinician Key Setup Guide, User Privacy Manual.